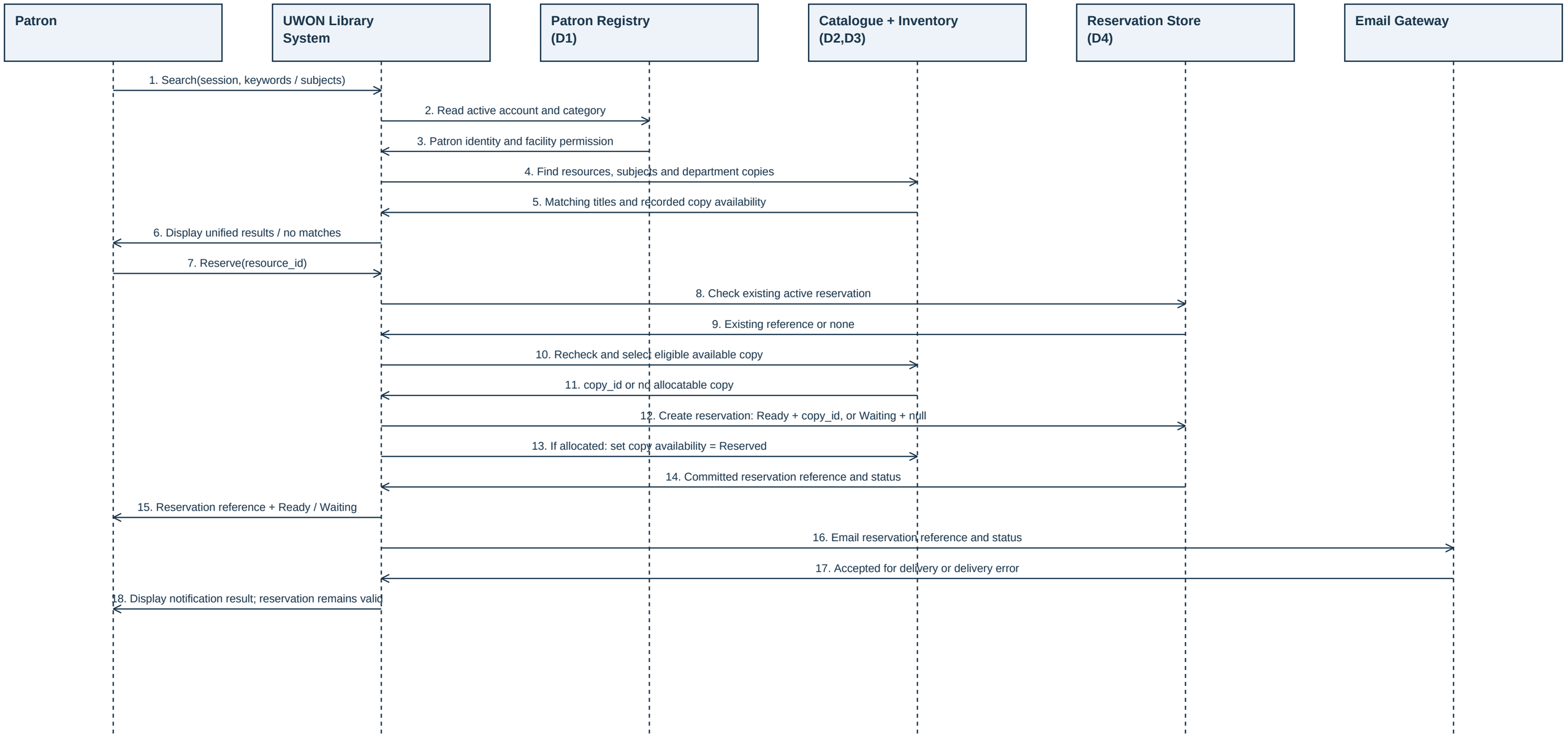
S1 — Event Trace Diagram

Search the unified catalogue and reserve a resource

An active, authorized patron searches across UWON department holdings using the shared subject vocabulary, reviews recorded availability, and reserves a resource. If an eligible copy is available, the same transaction allocates it and marks the reservation Ready; otherwise the reservation is Waiting without an allocated copy. The patron receives a reference and an email notification attempt, while duplicate active reservations return the existing reference.



Scenario conditions and alternatives

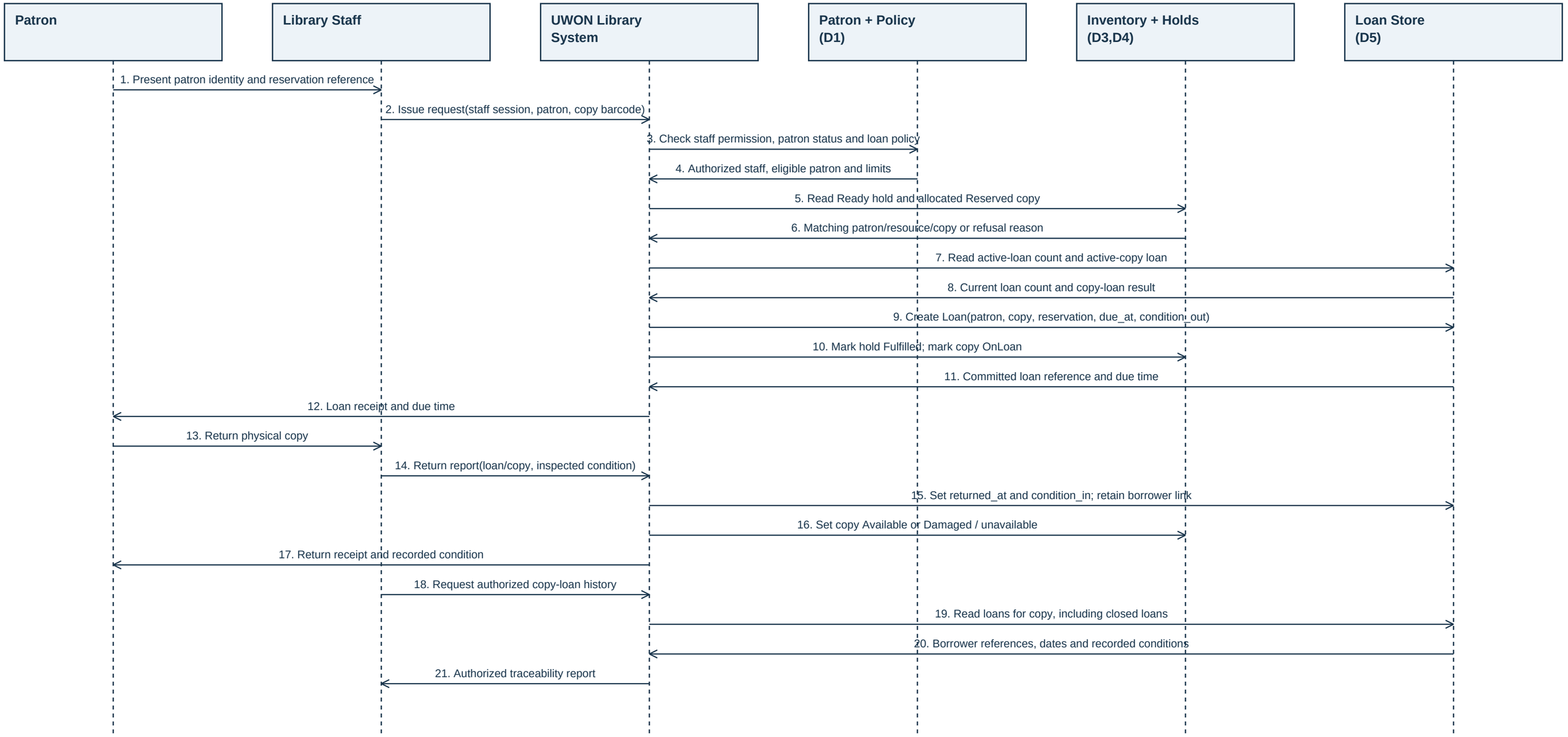
Precondition: registered active patron and facility permission. Reservation creation and any copy allocation are one atomic transaction; copies marked Binding, Missing, Damaged or OnLoan cannot be allocated. Alternatives: invalid permission -> deny without writes; no search match -> empty results; no available copy -> Waiting; duplicate hold -> existing reference; email failure does not cancel a committed reservation.

Time proceeds downward. Logical repositories refer to the same persistent data groups used in the ER and DFD views.

S2 — Event Trace Diagram

Borrow a reserved copy, return it and preserve borrower history

A patron presents a Ready reservation, and authorized library staff scan its allocated copy. The system checks account status, the applicable loan policy and reservation ownership before recording the loan and due time. On return, staff record the copy condition, the loan is closed without deleting borrower history, and the copy becomes Available or is withheld as Damaged.



Scenario conditions and alternatives

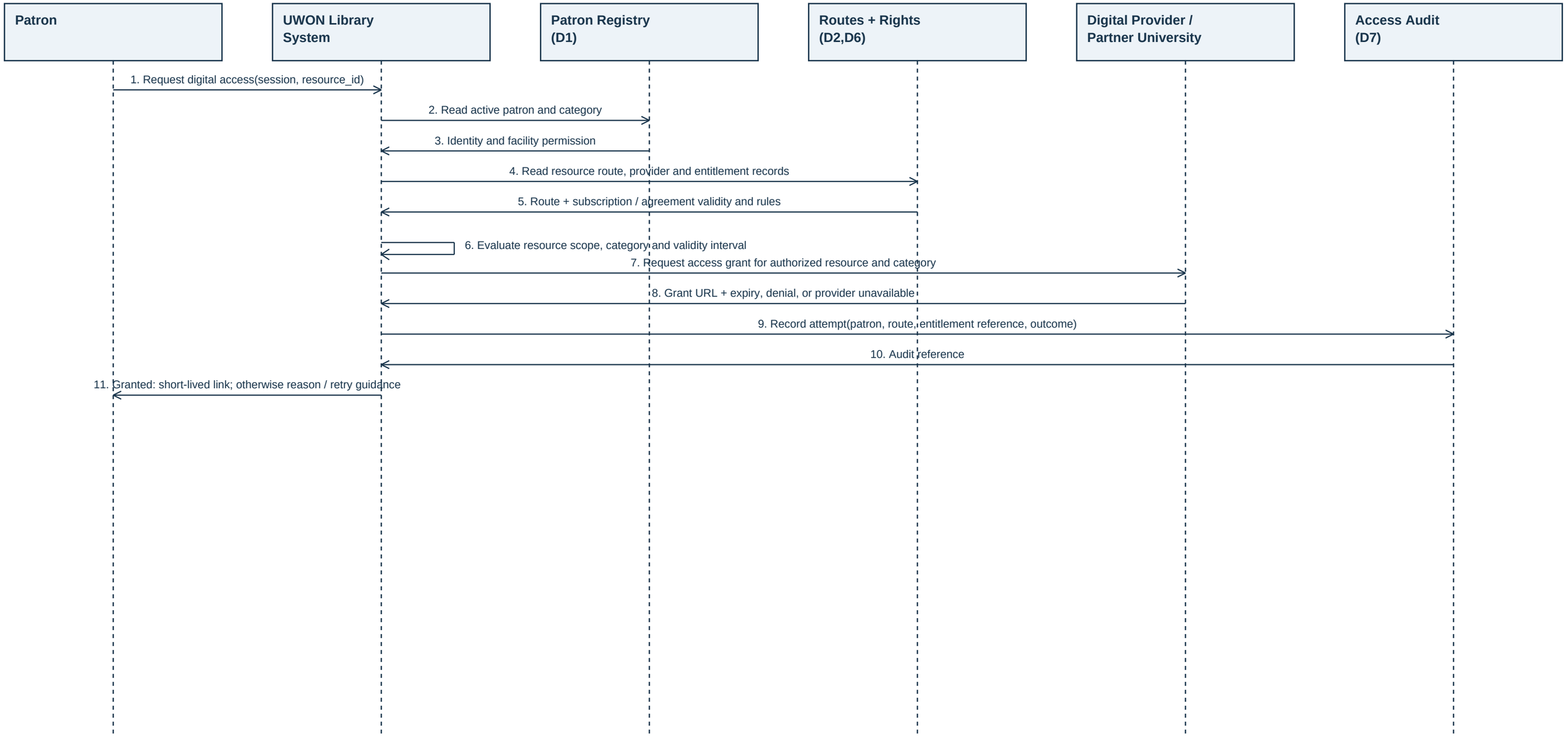
Preconditions: active patron, authorized staff, Ready reservation with allocated copy, and a known loan policy. Loan creation, hold fulfillment and copy status change are atomic; return updates are atomic and closed loans are retained. Alternatives: over limit, wrong patron, missing physical copy or unavailable copy -> refuse before issue; duplicate return -> existing return receipt; damage history supports investigation but does not by itself prove who caused damage.

Time proceeds downward. Logical repositories refer to the same persistent data groups used in the ER and DFD views.

S3 — Event Trace Diagram

Access a licensed e-journal or partner digital resource

An authenticated patron selects a digital resource and requests access. The system checks the patron category and either an active subscription or an applicable partner-university agreement, then asks the provider for an access grant. A valid provider response produces a short-lived access link and an audit record; denied or unavailable access is recorded without exposing protected content.



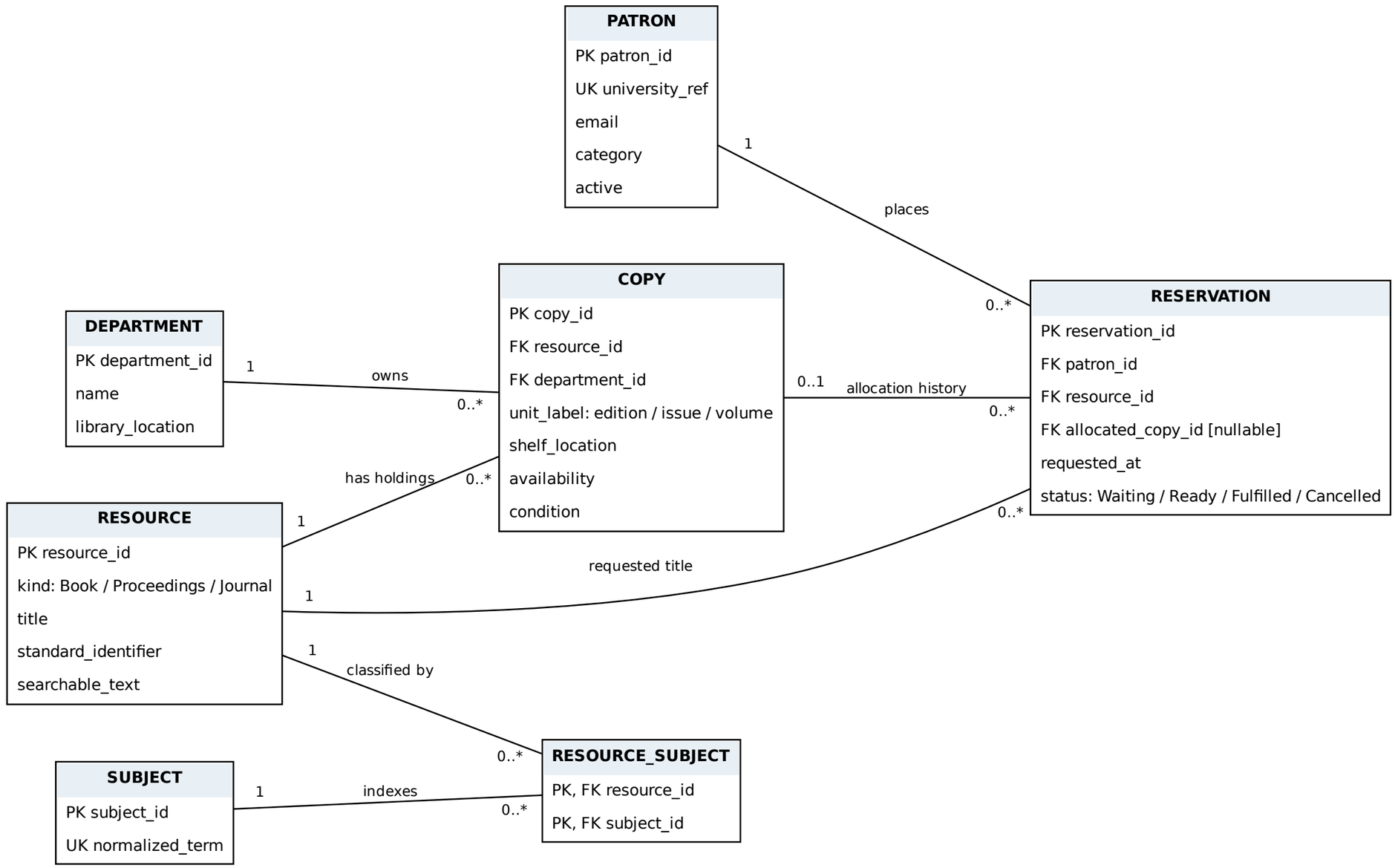
Scenario conditions and alternatives

Precondition: a selected digital catalogue record and active registered patron. Grant requires an active, in-scope Subscription OR Agreement plus provider acceptance; no entitlement means deny before the provider grant request. Each entitlement/provider attempt is retained with Granted, Denied or ProviderUnavailable; identity/facility denials stop before an attempt is created; a granted attempt references exactly one entitlement source, and the audit does not store a reusable access token.

Time proceeds downward. Logical repositories refer to the same persistent data groups used in the ER and DFD views.

S1 — Entity Relationship Diagram

Search the unified catalogue and reserve a resource

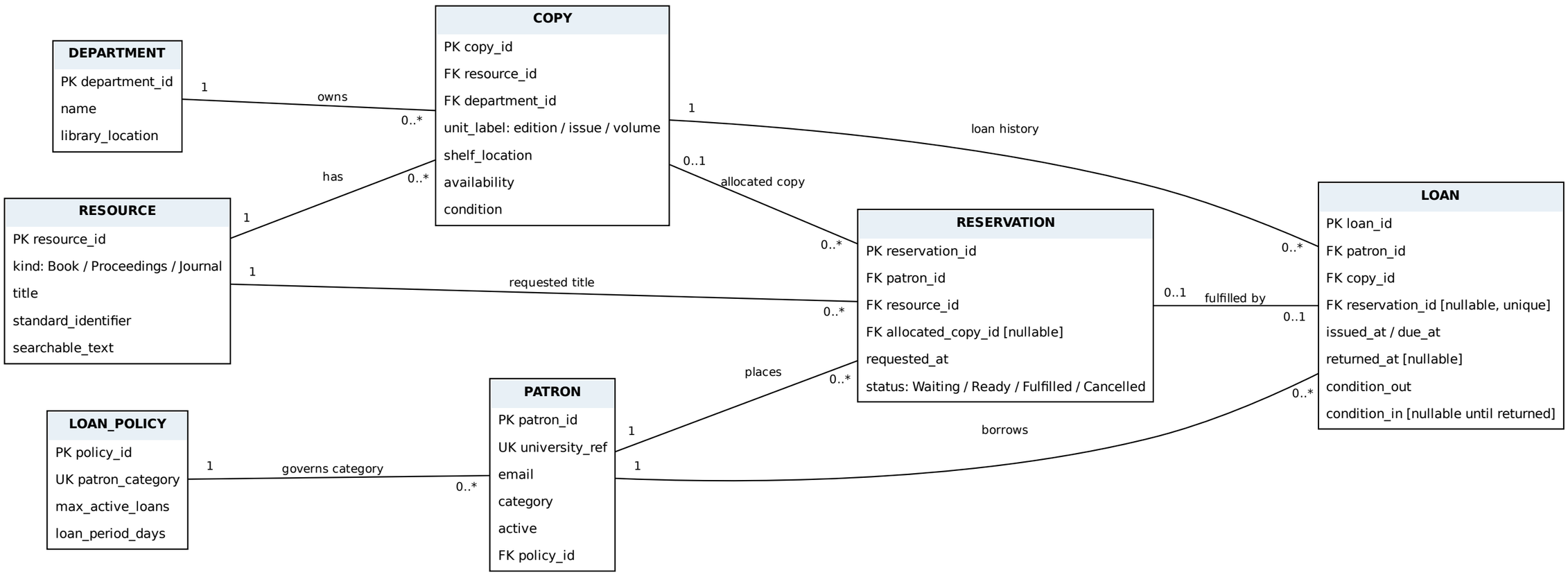


PK = primary key; FK = foreign key; UK = unique key. Numbers at an entity end show its allowed multiplicity for one entity at the opposite end.

Each Reservation belongs to exactly one Patron and Resource; allocated_copy_id is null while Waiting and required while Ready. Any allocated copy must belong to the same Resource. One copy has at most one active Ready allocation; only an eligible Available copy can become Reserved. RESOURCE_SUBJECT has a composite primary key; subjects are shared across departments.

S2 — Entity Relationship Diagram

Borrow a reserved copy, return it and preserve borrower history

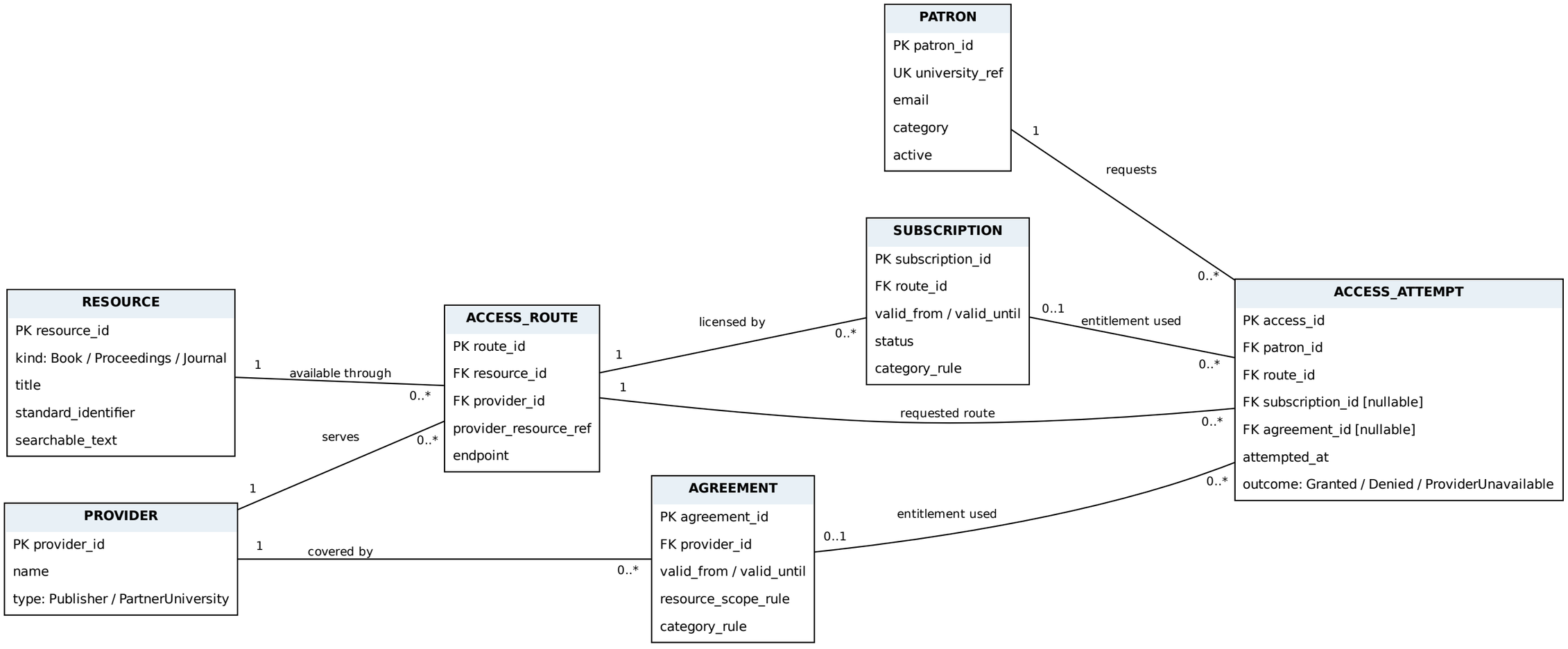


PK = primary key; FK = foreign key; UK = unique key. Numbers at an entity end show its allowed multiplicity for one entity at the opposite end.

In this view PATRON also has FK policy_id -> LOAN_POLICY. A Loan always links one Patron to one Copy; an optional reservation_id is unique and, when supplied, must match the same patron and allocated copy. At most one Loan per Copy can have returned_at null. Keep closed loans and condition records; return does not delete history. Copy becomes Available only if inspection permits circulation.

S3 — Entity Relationship Diagram

Access a licensed e-journal or partner digital resource

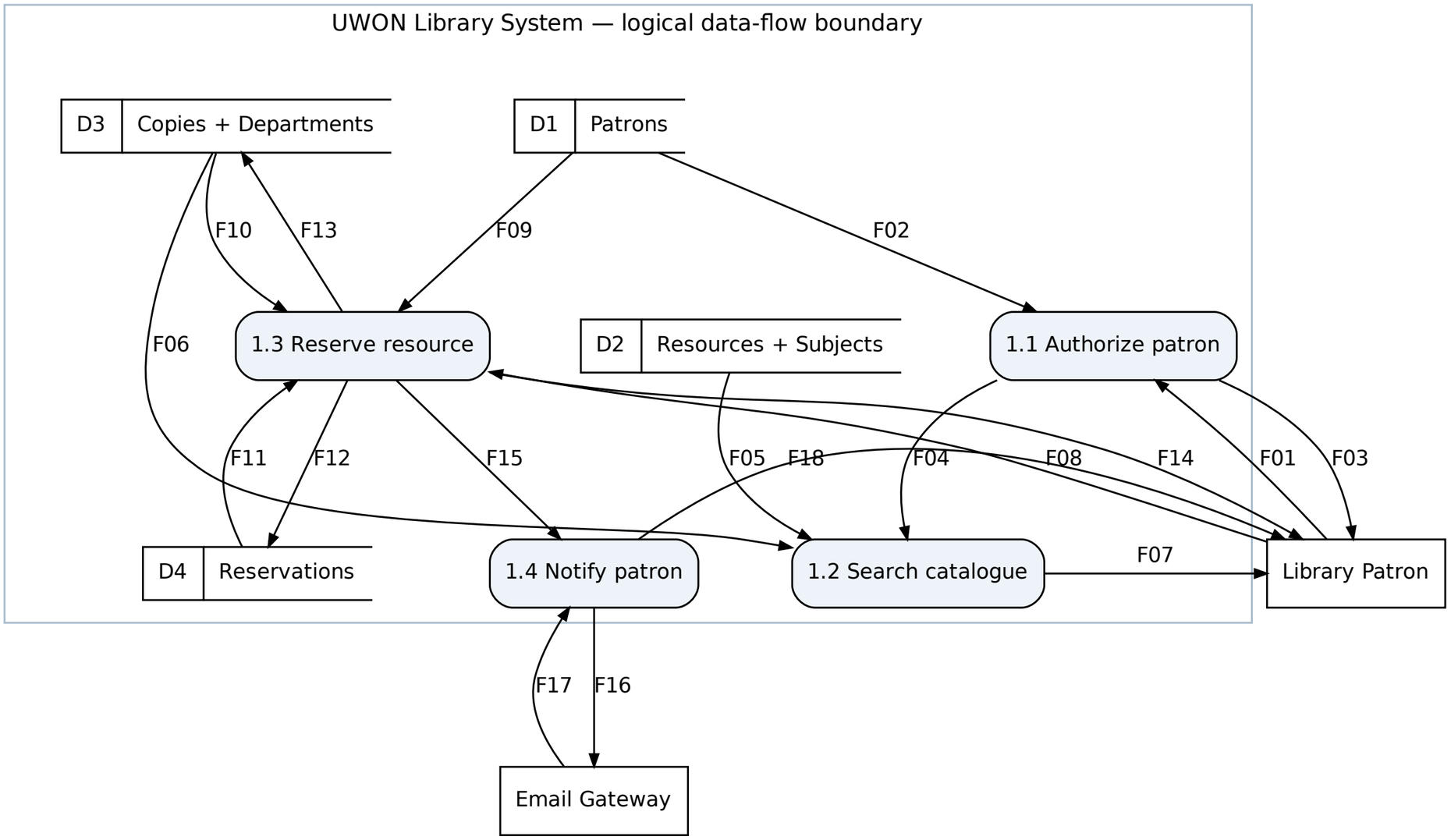


PK = primary key; FK = foreign key; UK = unique key. Numbers at an entity end show its allowed multiplicity for one entity at the opposite end.

For Granted attempts, exactly one of subscription_id or agreement_id is present (XOR); its route/provider, scope, category rule and validity interval must match the requested access. A Denied attempt may have neither reference. Partner access is represented by PROVIDER.type = PartnerUniversity plus an applicable Agreement. A grant depends on provider acceptance; audit stores outcome and references, not reusable grant tokens.

S1 — Dataflow Diagram

Search the unified catalogue and reserve a resource



Rectangles = external entities; rounded numbered boxes = processes; open-ended D# boxes = stores; F# arrows identify named data in the following flow dictionary, not time sequence.

D2 maps RESOURCE, SUBJECT and RESOURCE_SUBJECT; D3 maps COPY and DEPARTMENT; D4 maps RESERVATION. Process 1.3 performs copy recheck plus reservation/copy update atomically. D1 is read for every restricted operation. Email failure affects notification status, not the committed hold.

S1 — Dataflow Dictionary

Search the unified catalogue and reserve a resource

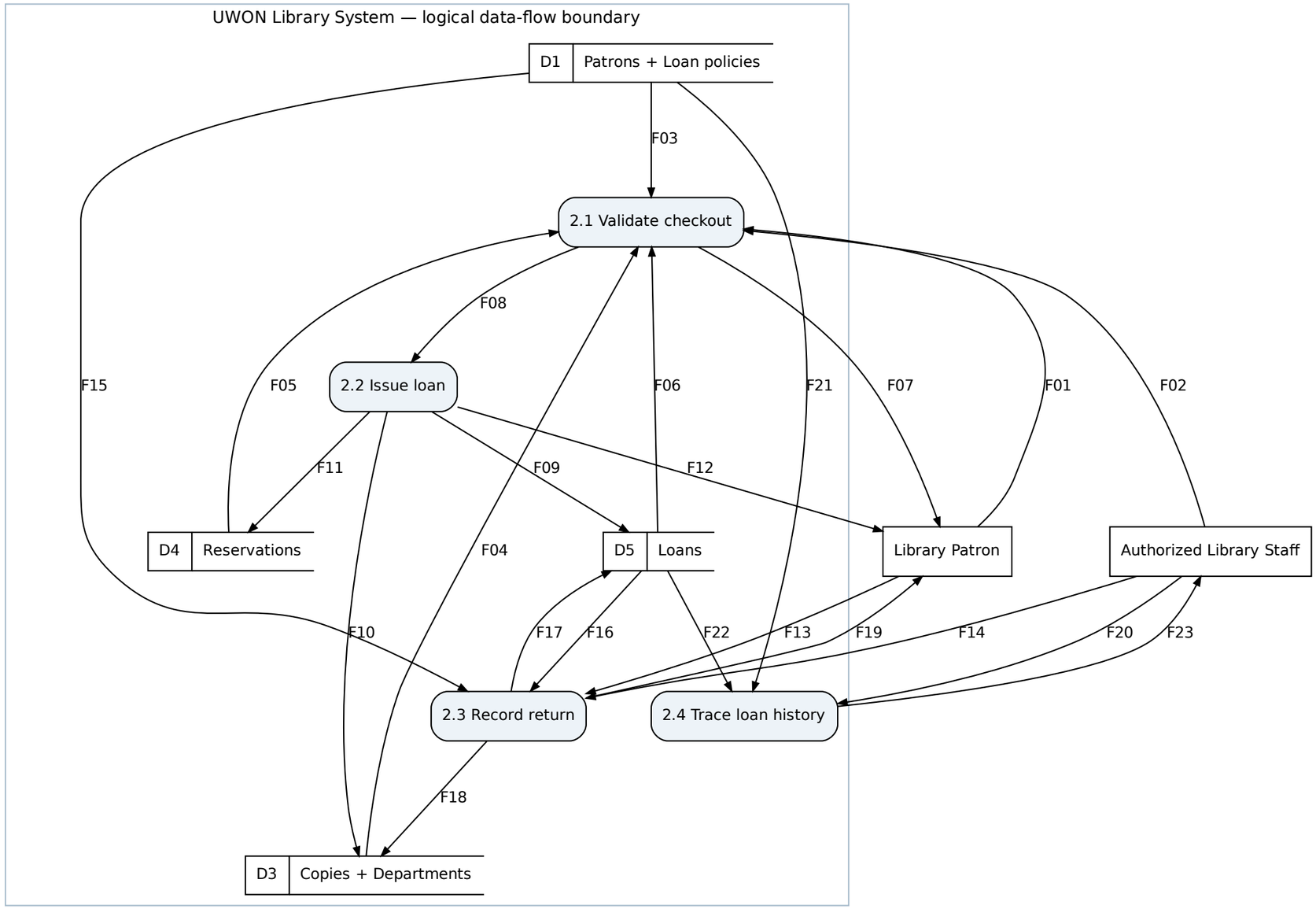
Each F-number on the preceding diagram identifies the named data flow listed below; it does not indicate execution order.

FLOW	FROM	TO	DATA
F01	Library Patron	1.1 Authorize patron	session + search criteria
F02	D1 Patrons	1.1 Authorize patron	active identity + category
F03	1.1 Authorize patron	Library Patron	authorization denial
F04	1.1 Authorize patron	1.2 Search catalogue	authorized search criteria
F05	D2 Resources + Subjects	1.2 Search catalogue	resource + subject records
F06	D3 Copies + Departments	1.2 Search catalogue	holding locations + availability
F07	1.2 Search catalogue	Library Patron	unified search results
F08	Library Patron	1.3 Reserve resource	selected resource + session
F09	D1 Patrons	1.3 Reserve resource	identity + category
F10	D3 Copies + Departments	1.3 Reserve resource	eligible copy state
F11	D4 Reservations	1.3 Reserve resource	existing active reservations
F12	1.3 Reserve resource	D4 Reservations	new reservation record
F13	1.3 Reserve resource	D3 Copies + Departments	allocated-copy Reserved state
F14	1.3 Reserve resource	Library Patron	Ready / Waiting reference or refusal
F15	1.3 Reserve resource	1.4 Notify patron	committed hold + recipient data
F16	1.4 Notify patron	Email Gateway	reservation email
F17	Email Gateway	1.4 Notify patron	delivery acceptance / error
F18	1.4 Notify patron	Library Patron	notification status

D2 maps RESOURCE, SUBJECT and RESOURCE_SUBJECT; D3 maps COPY and DEPARTMENT; D4 maps RESERVATION. Process 1.3 performs copy recheck plus reservation/copy update atomically. D1 is read for every restricted operation. Email failure affects notification status, not the committed hold.

S2 — Dataflow Diagram

Borrow a reserved copy, return it and preserve borrower history



Rectangles = external entities; rounded numbered boxes = processes; open-ended D# boxes = stores; F# arrows identify named data in the following flow dictionary, not time sequence.

D1 maps PATRON and LOAN_POLICY; D3 maps COPY/DEPARTMENT; D4 maps RESERVATION; D5 maps LOAN. Issue and return updates each form an atomic unit. Patron and Staff are external roles, not data stores; recorded patron identity remains in D1 and loan links in D5. Read-only history access is restricted to authorized staff.

S2 — Dataflow Dictionary

Borrow a reserved copy, return it and preserve borrower history

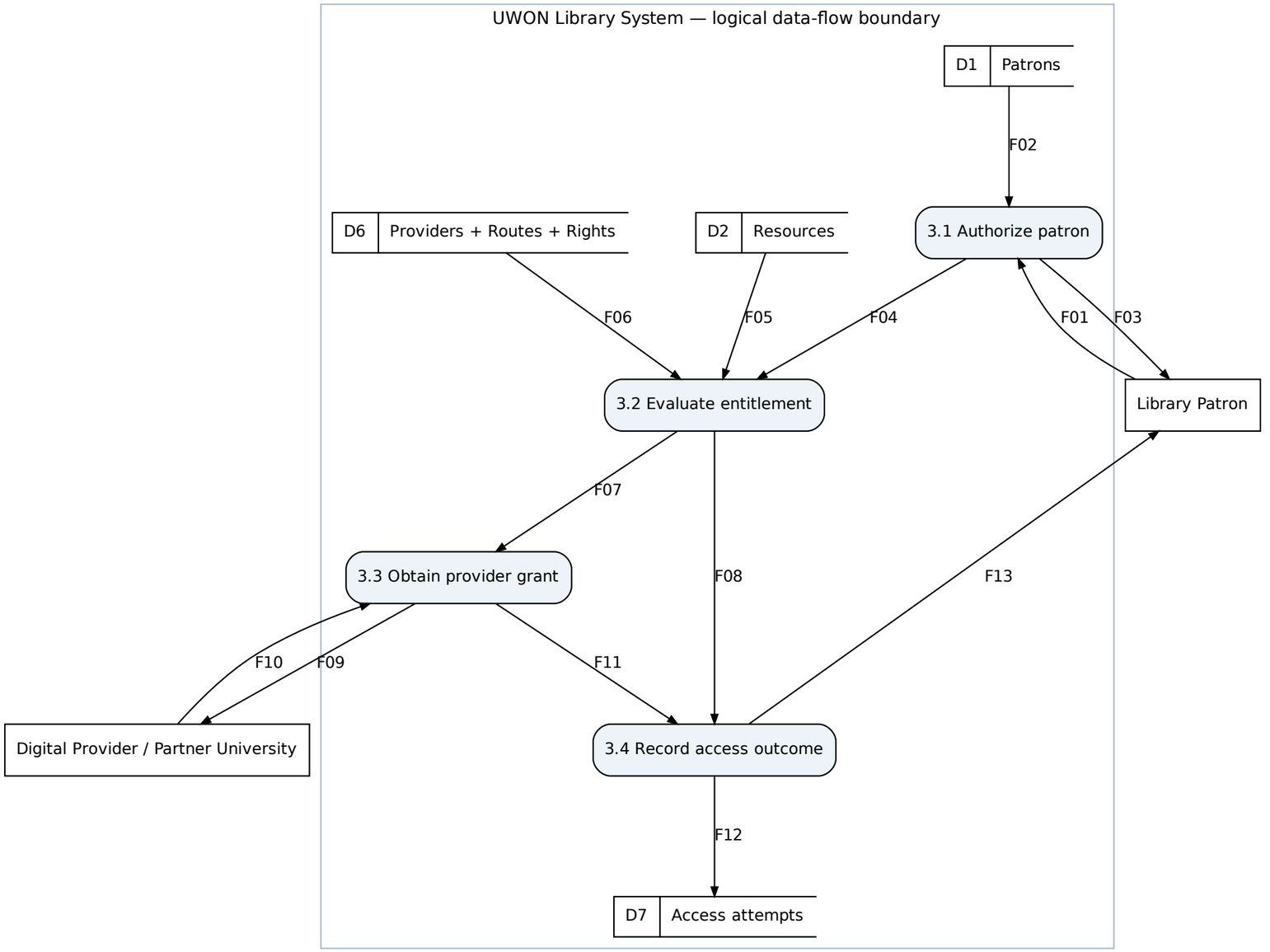
Each F-number on the preceding diagram identifies the named data flow listed below; it does not indicate execution order.

FLOW	FROM	TO	DATA
F01	Library Patron	2.1 Validate checkout	patron identity + hold reference
F02	Authorized Library Staff	2.1 Validate checkout	staff session + copy barcode
F03	D1 Patrons + Loan policies	2.1 Validate checkout	account + category + loan rules
F04	D3 Copies + Departments	2.1 Validate checkout	copy availability + condition
F05	D4 Reservations	2.1 Validate checkout	Ready reservation + allocation
F06	D5 Loans	2.1 Validate checkout	active loan count + copy loans
F07	2.1 Validate checkout	Library Patron	eligibility / limit refusal
F08	2.1 Validate checkout	2.2 Issue loan	validated patron + copy + hold
F09	2.2 Issue loan	D5 Loans	new Loan + due time + condition_out
F10	2.2 Issue loan	D3 Copies + Departments	copy OnLoan state
F11	2.2 Issue loan	D4 Reservations	Fulfilled reservation state
F12	2.2 Issue loan	Library Patron	loan receipt + due time
F13	Library Patron	2.3 Record return	returned copy / loan reference
F14	Authorized Library Staff	2.3 Record return	staff session + inspected condition
F15	D1 Patrons + Loan policies	2.3 Record return	staff authorization
F16	D5 Loans	2.3 Record return	active / already closed loan
F17	2.3 Record return	D5 Loans	returned_at + condition_in
F18	2.3 Record return	D3 Copies + Departments	Available / Damaged copy state
F19	2.3 Record return	Library Patron	return receipt
F20	Authorized Library Staff	2.4 Trace loan history	authorized copy-history criteria
F21	D1 Patrons + Loan policies	2.4 Trace loan history	staff authorization
F22	D5 Loans	2.4 Trace loan history	retained loan history
F23	2.4 Trace loan history	Authorized Library Staff	borrower / condition history report

D1 maps PATRON and LOAN_POLICY; D3 maps COPY/DEPARTMENT; D4 maps RESERVATION; D5 maps LOAN. Issue and return updates each form an atomic unit. Patron and Staff are external roles, not data stores; recorded patron identity remains in D1 and loan links in D5. Read-only history access is restricted to authorized staff.

S3 — Dataflow Diagram

Access a licensed e-journal or partner digital resource



Rectangles = external entities; rounded numbered boxes = processes; open-ended D# boxes = stores; F# arrows identify named data in the following flow dictionary, not time sequence.

D6 maps PROVIDER, ACCESS_ROUTE, SUBSCRIPTION and AGREEMENT; D7 maps ACCESS_ATTEMPT. Only eligible requests reach the provider; granted access also requires provider acceptance. Attempt records preserve outcome and selected entitlement source, while reusable access tokens are not persisted. No entitlement or provider denial produces no content link.

S3 — Dataflow Dictionary

Access a licensed e-journal or partner digital resource

Each F-number on the preceding diagram identifies the named data flow listed below; it does not indicate execution order.

FLOW	FROM	TO	DATA
F01	Library Patron	3.1 Authorize patron	session + resource request
F02	D1 Patrons	3.1 Authorize patron	active identity + category
F03	3.1 Authorize patron	Library Patron	identity / facility denial
F04	3.1 Authorize patron	3.2 Evaluate entitlement	authorized patron + resource
F05	D2 Resources	3.2 Evaluate entitlement	digital resource metadata
F06	D6 Providers + Routes + Rights	3.2 Evaluate entitlement	route + provider + subscription / agreement
F07	3.2 Evaluate entitlement	3.3 Obtain provider grant	eligible access context
F08	3.2 Evaluate entitlement	3.4 Record access outcome	denied entitlement outcome
F09	3.3 Obtain provider grant	Digital Provider / Partner University	resource + authorized access context
F10	Digital Provider / Partner University	3.3 Obtain provider grant	grant URL + expiry / denial / unavailable
F11	3.3 Obtain provider grant	3.4 Record access outcome	provider outcome + entitlement reference
F12	3.4 Record access outcome	D7 Access attempts	access-attempt audit record
F13	3.4 Record access outcome	Library Patron	temporary link or reason / retry guidance

D6 maps PROVIDER, ACCESS_ROUTE, SUBSCRIPTION and AGREEMENT; D7 maps ACCESS_ATTEMPT. Only eligible requests reach the provider; granted access also requires provider acceptance. Attempt records preserve outcome and selected entitlement source, while reusable access tokens are not persisted. No entitlement or provider denial produces no content link.